MAINTENANCE TROUBLESHOOTING

Line Down: 50 Checks Before You Call for Help

A pocket checklist for maintenance technicians and controls engineers. Work top to bottom: most stoppages are found in the first few checks. Written by a controls engineer with 30 years on the plant floor.

Safety first. Follow your plant's lockout/tagout procedure and verify zero energy before touching anything inside a panel or a machine. Wear the PPE the arc-flash label and your plant rules require. Only qualified people should take live measurements. Never bypass a safety device to get a line running.

1 · Before you touch anything

- [ ] Talk to the operator: what happened just before it stopped, and has it happened before?
- [ ] Read every active alarm on the HMI and write down the first one: the first alarm is usually the cause, the rest are side effects.
- [ ] Note any fault code on drives, PLC modules or safety relays **before** you reset anything.
- [ ] Look, listen and smell: tripped breakers, burnt odor, loose guards, product jams, air leaks.
- [ ] Check the machine's history log or journal for the same symptom and its last fix.

2 · Machine won't start

- [ ] All E-stops pulled out, and the safety relay or safety PLC reset (look at its status LEDs).
- [ ] Guard and door switches made; light curtains clear.
- [ ] Control power present: 24 VDC supply output measured, not assumed.
- [ ] Master control relay (MCR) energized.
- [ ] PLC in RUN mode, with no major fault or flashing fault LED.
- [ ] Every start permissive met: pressure, level, temperature, position, upstream and downstream ready.
- [ ] Main disconnect on and motor overload relay not tripped.
- [ ] Drive shows Ready, not Fault or Not Enabled.
- [ ] Machine in the mode it needs: Auto/Manual, Local/Remote.
- [ ] Recipe or setpoints loaded and within limits.

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3 · Motors and VFDs

- [ ] Record the drive's fault code and fault history before resetting; look it up in the drive manual.
- [ ] Measure incoming voltage phase to phase on all three phases; they should be balanced and within the drive's rating.
- [ ] Overcurrent faults on start: acceleration time too short, mechanical bind, or a load that is too heavy. Turn the shaft by hand with power locked out.
- [ ] Overvoltage (DC bus) faults while stopping: deceleration time too short for the load's inertia, or a braking resistor is needed or has failed.
- [ ] Ground fault: suspect the motor cable or motor insulation. Insulation-test (megger) the motor and cable **with the drive disconnected**, never through the drive.
- [ ] Motor overload settings match the nameplate full-load amps (FLA).
- [ ] Drive cooling fans run and heatsinks are clean; overtemperature faults often trace back to dust.
- [ ] Check the coupling, belts, bearings and gearbox before blaming the electrical side.

4 · Discrete sensors

- [ ] Does the sensor's own LED change when the target is present? Then check the PLC input LED for the same point.
- [ ] Sensor LED changes but the PLC input does not: check the wiring, the input card's common and fuse, and PNP (sourcing) versus NPN (sinking) wiring.
- [ ] Neither LED changes: check sensor power, sensing distance, target material and alignment, and a dirty lens or reflector.
- [ ] Look for crushed or chafed cables and loose quick-disconnect connectors at moving parts.
- [ ] Replacing a sensor? Match the output type, the sensing range and the connector.

5 · Pneumatics

- [ ] Supply pressure at the machine regulator, not only at the compressor.
- [ ] Solenoid valve coil voltage present and the valve LED lit when the output is on.
- [ ] Use the valve's manual override to tell an electrical fault from a mechanical one.
- [ ] Blocked exhaust or muffler, kinked tubing, or flow controls closed down too far.
- [ ] Water or oil in the lines: drain the filters; check the dryer.
- [ ] Cylinder drifting or slow under load: worn piston seals letting air bypass.

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6 · 4–20 mA loops and instruments

- [ ] Measure the loop current in series with a loop calibrator or a meter on its mA range (or across the transmitter's test terminals).
- [ ] 4 mA is the bottom of the range and 20 mA the top. Below about 3.8 mA or above about 20.5 mA usually means a failed transmitter, open wire or short (NAMUR NE 43 failure signals are ≤ 3.6 mA and ≥ 21 mA).
- [ ] 0 mA: open loop, blown fuse, missing loop power, or loose terminal.
- [ ] Check the loop power supply voltage and that the total loop resistance is within the transmitter's limit.
- [ ] PLC reading wrong but the current is right? The scaling in the PLC must match the transmitter's calibrated range.
- [ ] Readings jumping: shield grounded at one end only, and signal cables kept away from VFD output cables.

7 · PLC and network communication

- [ ] Link and activity LEDs on the PLC, the switch port and the device.
- [ ] Ping the device from a laptop on the same subnet; no reply means cable, switch, power or a wrong IP address.
- [ ] Replaced a drive, I/O block or HMI? The new one must be given the same IP address and settings as the old one.
- [ ] Two devices with the same IP address cause connections that drop in and out.
- [ ] Check the I/O module or adapter status in the PLC software (I/O tree) for faulted connections.
- [ ] Look at cables and connectors at doors, cable tracks and anywhere that moves or vibrates.

8 · After the fix

- [ ] Write down the symptom, the cause, the fix and the parts used, by machine, with the date and time.
- [ ] Tell the next shift what happened and what to watch for.
- [ ] Ask: could a PM task, a spare part on the shelf or an alarm have caught this sooner? Add it.
- [ ] Restock any spare you used.

The fastest fix is the one somebody already found. Keep a searchable record of every fix by machine, and the next technician starts from the answer, not from zero.

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WORKSHEET

What does one hour of downtime cost you?

Fill in your own numbers. Knowing this figure makes it easy to decide which fixes, spares and tools pay for themselves.

Item	How to work it out	Your number
A · Lost production	Units per hour × profit per unit (or sales value, if you can't make it up later)	$
B · Idle labor	People on the line × their hourly cost, including benefits	$
C · Scrap and restart	Product lost at the stop and on restart, plus cleanup	$
D · Recovery overtime	Overtime hours needed to catch up × overtime rate	$
E · Repair	Technician time, call-out fees, rush parts and freight	$
Cost of one hour	**A + B + C + D + E**	**$**
Unplanned downtime hours per month	From your logs or your best estimate	
Cost per year	**Cost of one hour × hours per month × 12**	**$**

Example: $2,000 an hour × 4 hours a month × 12 = **$96,000 a year**. Cutting that by just 10% saves $9,600 a year.

Tools that help, from Syon Automation LLC

PlantProof: SCADA, historian and maintenance in one install · Datumline: field journal and time clock that logs every fix by machine · RungWise PLC Academy and TechProof: training your techs on the plant floor · AlertPush: PLC alarms pushed to the right phone

Free trials and savings calculators: syonautomation.com · After-hours PLC and VFD help: text 470-228-2167

This guide is general information to help qualified people troubleshoot. It does not replace your equipment manuals, your plant's safety procedures or the training required for electrical work. Syon Automation LLC accepts no liability for how it is used.